

# Keplerian turbulence & planetesimal formation

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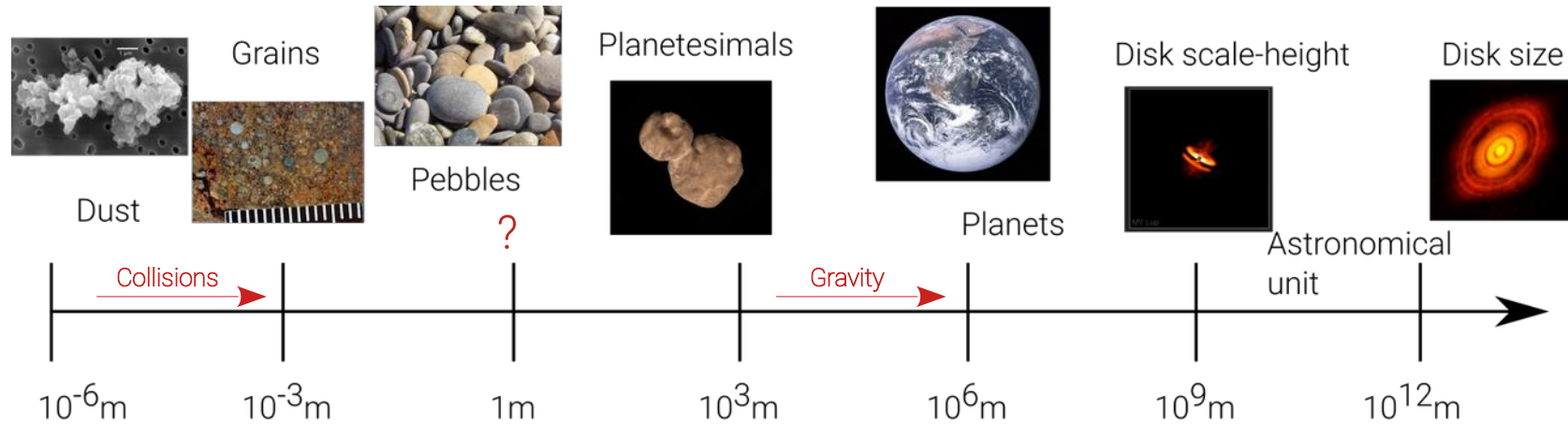
4 INRIA, Team Callisto



PFITS+



# Barriers to planetesimal formation



- Growth barriers: bouncing, fragmentation
  - ↳ Concentration of solids: Rossby wave instability (vortensity extremum), streaming instability (high dust-to-gas ratio), Convective over stability – Vertical shear instability – Zombie vortex instability (thermal processes)...
- Drift barrier
  - ↳ Pressure bumps

# A unique solution ?

- Turbulence

- ↳ A known process in protoplanetary disks

- Debated intensity
    - Probably of low intensity

- ↳ Previously thought to oppose to grain concentration

- ↳ Previously considered as increasing grain drift

(e.g. Fromang & Papaloizou, 2006, Youdin & Lithwick 2007)

- Models

- ↳ Mean-field approach : viscous ( $\alpha$ ) and diffusive coefficient

- ↳ Hydro/MHD turbulent cascade → homogeneous and isotropic turbulence

# Model

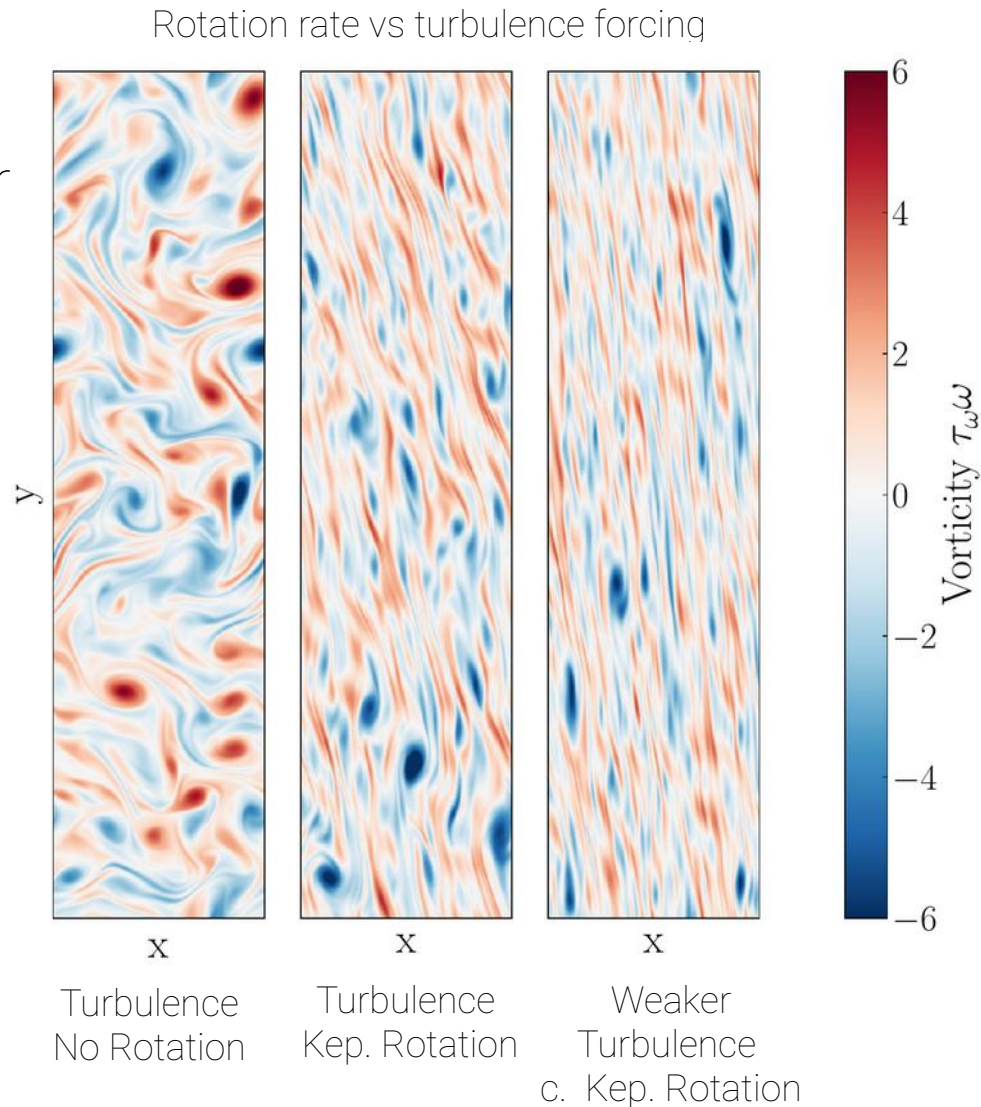
- Local: Shearing box – Rotation & shear rate

$$\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho_g} \nabla p + \nu \nabla^2 \mathbf{v} - 2\mathbf{\Omega} \times \mathbf{v} + 3\Omega^2 x \mathbf{e}_x$$

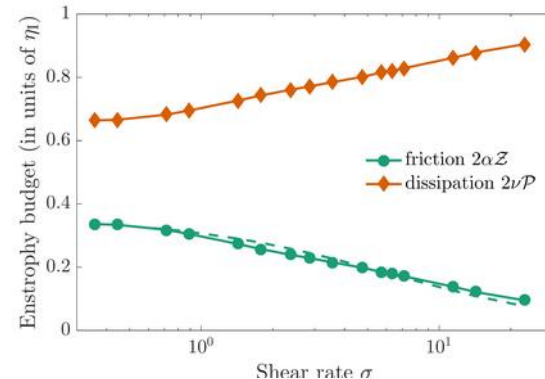
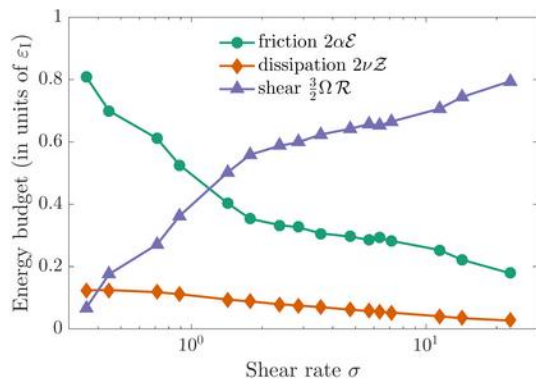
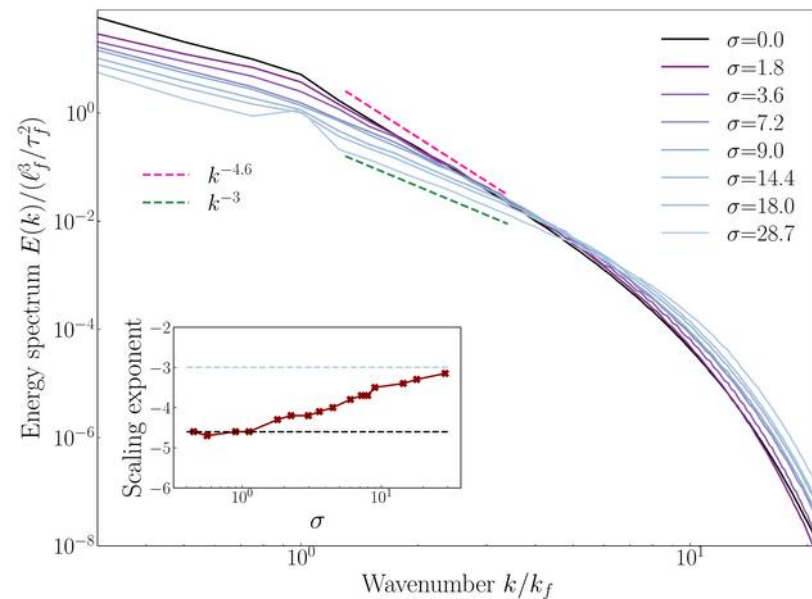
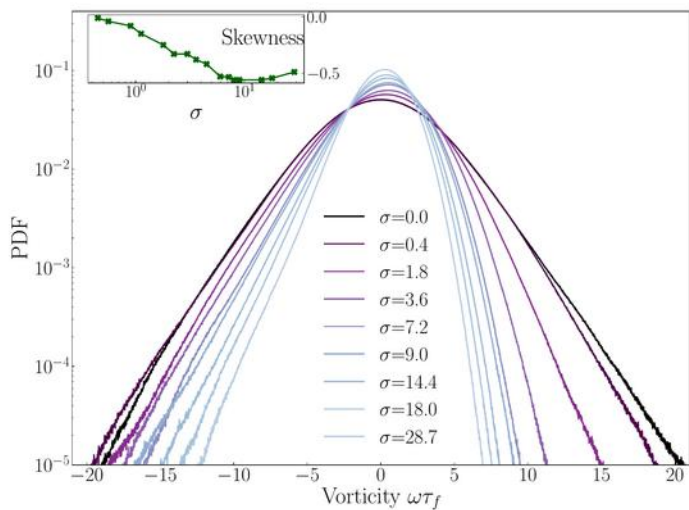
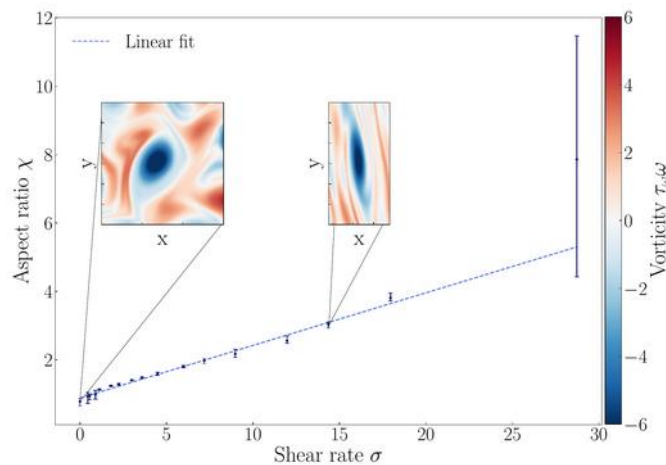
$$\nabla \cdot \mathbf{v} = 0$$

$$\sigma = \frac{3}{2} \Omega \tau_f$$

- Study of turbulence
  - ↳ Not transition to turbulence
  - ↳ Direct cascade to small scales
  - ↳ 2D forced incompressible turbulence
- Evolution of turbulence properties
  - ↳ Circular/elongated eddies (aspect ratio)
  - ↳ Cyclonic/anticyclonic eddies



# Keplerian turbulence



$$\begin{aligned}\frac{d\mathcal{E}}{dt} &= 0 = -2\nu\mathcal{Z} - 2\alpha\mathcal{E} - \frac{3}{2}\Omega\mathcal{R} + \varepsilon_1, \\ \frac{d\mathcal{Z}}{dt} &= 0 = -2\nu\mathcal{P} - 2\alpha\mathcal{Z} + \eta_1.\end{aligned}$$

$$\begin{aligned}\mathcal{Z} &= \frac{1}{2}\langle\omega^2\rangle & \mathcal{P} &= (1/2)\langle\|\nabla\omega\|^2\rangle \\ \mathcal{E} &= \frac{1}{2}\langle\|\mathbf{u}\|^2\rangle & \mathcal{R} &= \langle u_x u_y \rangle\end{aligned}$$

# Model

- Local: Shearing box – Rotation & shear rate

$$\partial_t \mathbf{v} + \mathbf{v} \cdot \nabla \mathbf{v} = -\frac{1}{\rho_g} \nabla p + \nu \nabla^2 \mathbf{v} - 2\mathbf{\Omega} \times \mathbf{v} + 3\Omega^2 x \mathbf{e}_x$$

$$\nabla \cdot \mathbf{v} = 0$$

$$\sigma = \frac{3}{2} \Omega \tau_f$$

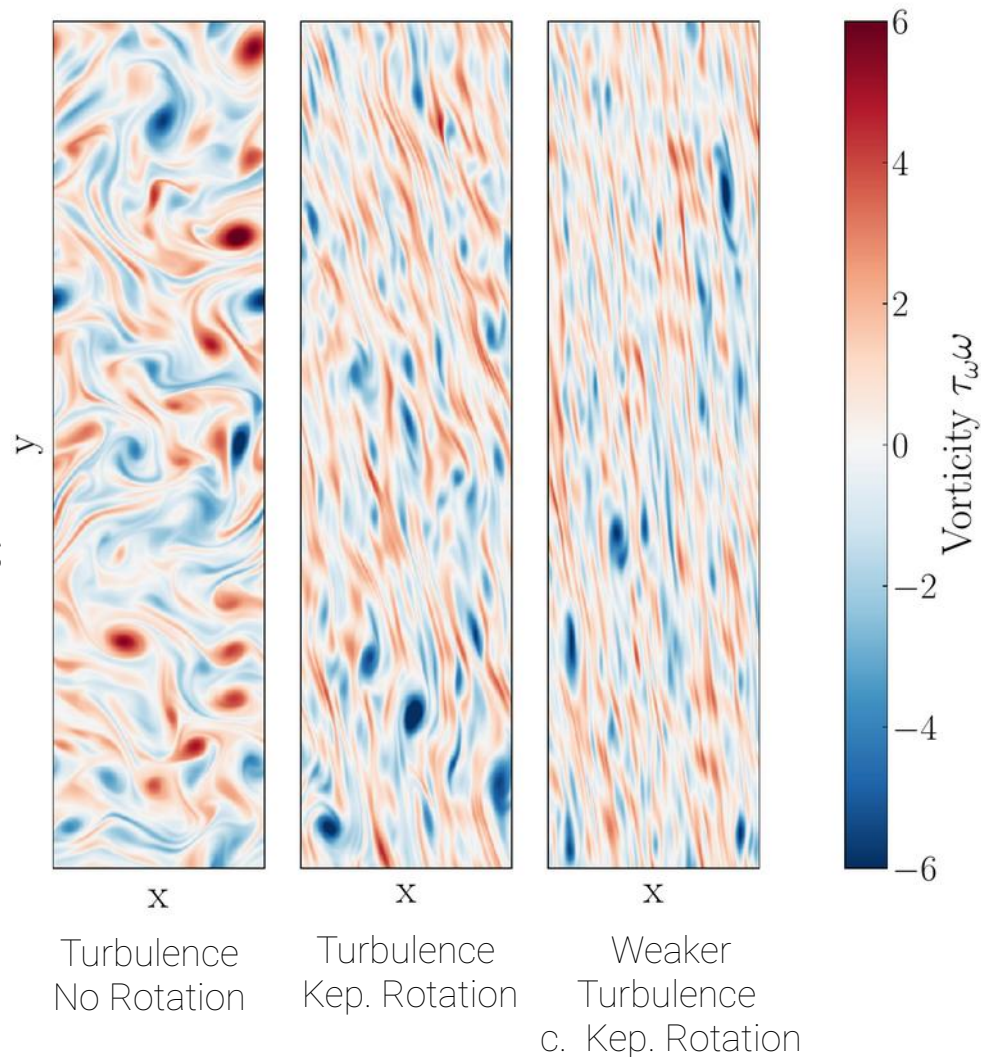
- Lagrangian populations of particles – Stopping time & Stokes number

$$\frac{d\mathbf{V}_p}{dt} = -\frac{1}{\tau_p} [\mathbf{V}_p - \mathbf{v}(\mathbf{R}_p, t)] - 2\mathbf{\Omega} \times \mathbf{V}_p + 3\Omega^2 x_p \mathbf{e}_x$$

$$St_\Omega = \tau_p \Omega$$

$$St_\omega = \tau_p / \tau_\omega = \langle \omega^2 \rangle^{1/2} \tau_p$$

Rotation rate vs turbulence forcing





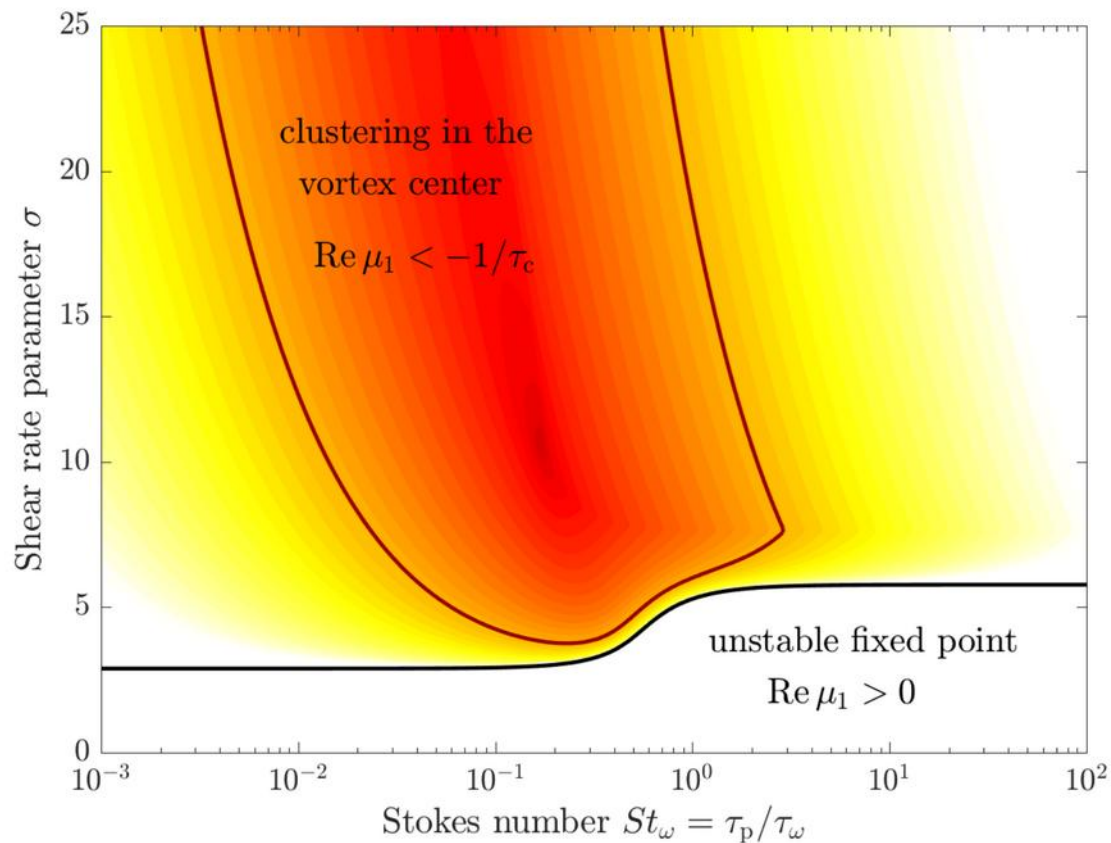
# Kida model for anticyclonic eddies

$$\mathbf{v} = \mathbb{A} \begin{bmatrix} x \\ y \end{bmatrix} \quad \mathbb{A} = \frac{3}{2} \frac{\Omega}{\chi - 1} \begin{bmatrix} 0 & \chi^{-1} \\ -\chi & 0 \end{bmatrix}$$

$$\frac{d}{dt} \begin{bmatrix} \mathbf{R}_p \\ \mathbf{V}_p \end{bmatrix} = \begin{bmatrix} \mathbb{0} & \mathbb{1} \\ \frac{1}{\tau_p} \mathbb{A} + 3\Omega^2 \mathbf{e}_{xx} & -\frac{1}{\tau_p} \mathbb{1} - 2\Omega \mathbb{J} \end{bmatrix} \begin{bmatrix} \mathbf{R}_p \\ \mathbf{V}_p \end{bmatrix}$$

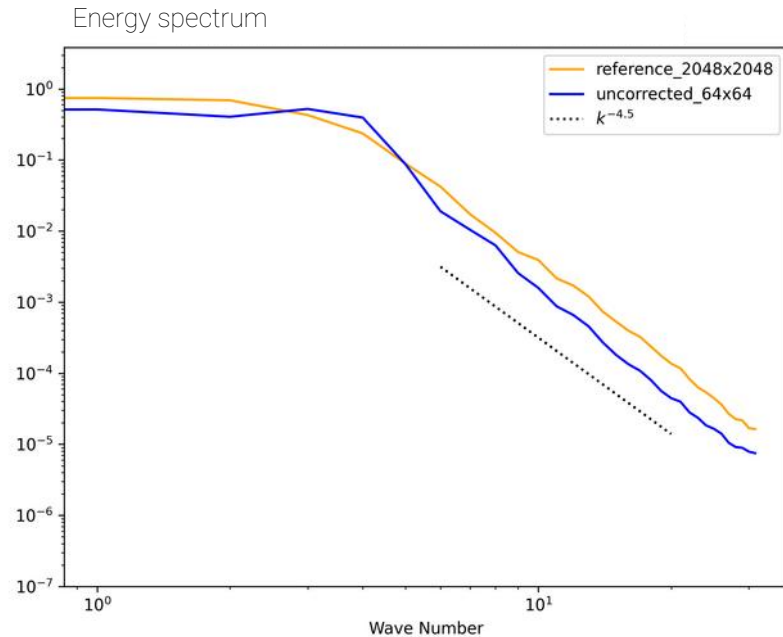
$$\mathbf{e}_{xx} = \begin{bmatrix} 1 & 0 & 0 & 0 \end{bmatrix} \quad \mathbb{J} = \begin{bmatrix} 0 & -1 & 1 & 0 \end{bmatrix}$$

Is Kida model a good approximation  
for Keplerian turbulence eddies?



# How to reduce computational cost of turbulence simulations?

- Reduce the resolution?



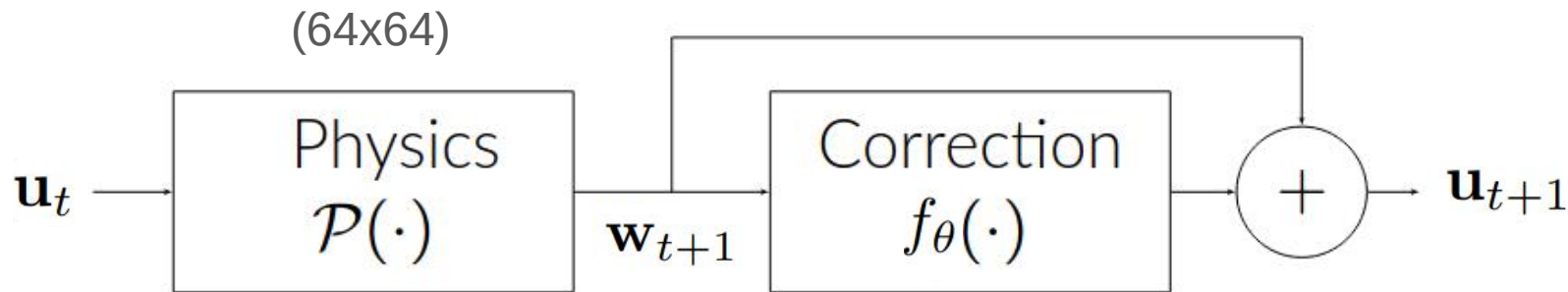
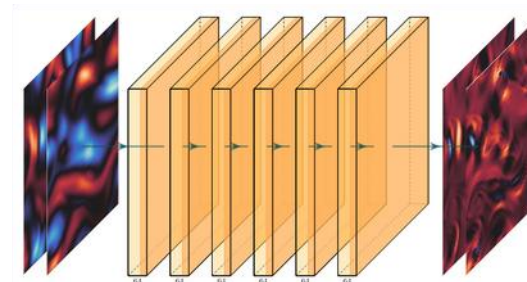
|          |             |
|----------|-------------|
| $2048^2$ | ~ 7 minutes |
| $64^2$   | ~ 2 seconds |



# How to reduce computational cost of turbulence simulations?

- Reduce the resolution
- Correct the error due to low resolution

$$\mathbf{u}_{t+1} = \mathcal{P}(\mathbf{u}_t) + f_{\theta}(\mathcal{P}(\mathbf{u}_t))$$



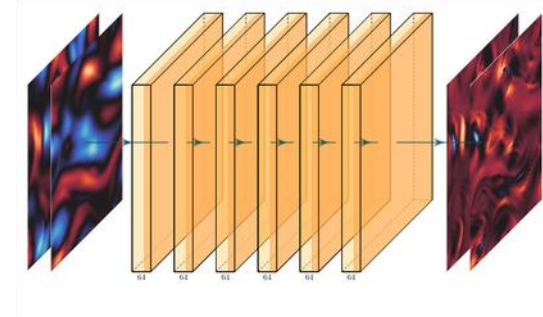
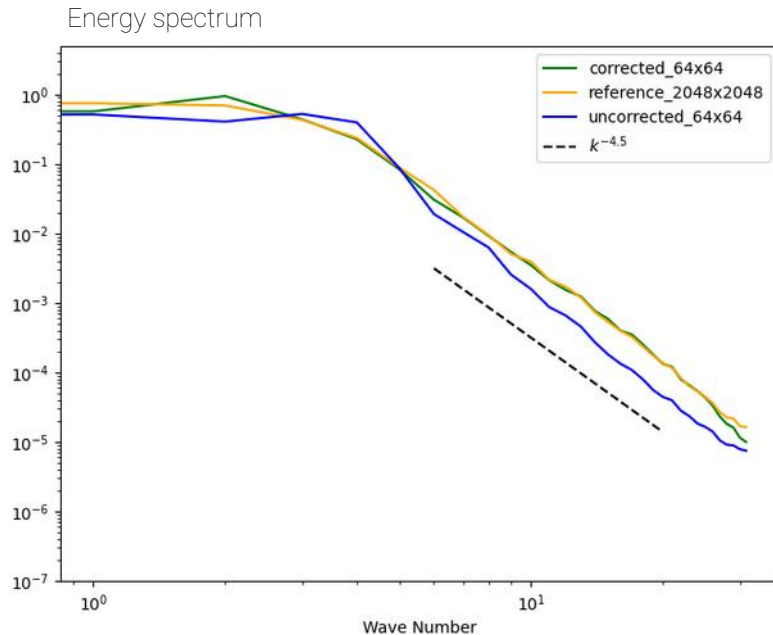
Numerical Navier-Stokes Solver

Neural Network Corrector

Amelie Lam PhD work  
Collaboration with André Ferrari  
& Sara El Bouch

# How to reduce computational cost of turbulence simulations?

- Reduce the resolution
- Correct the error due to low resolution



|                              |             |
|------------------------------|-------------|
| 2048 <sup>2</sup>            | ~ 7 minutes |
| 64 <sup>2</sup> + correction | ~ 6 seconds |